

Little Explorers · Class 6

Hot & Cold Temperature Light

Red means warm. Blue means cool!

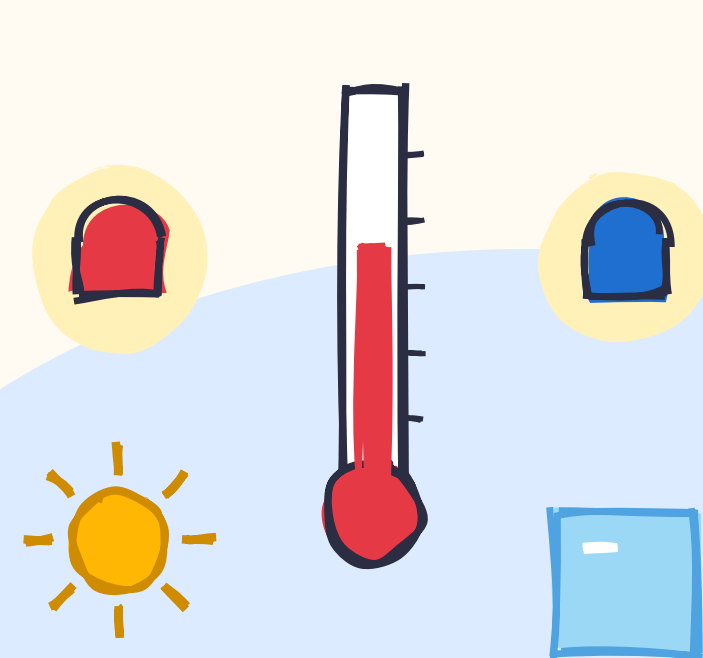
Temperature sensor

Red and blue lights

Choosing a number

Did you know?

Your body stays at about 37 degrees, even on a cold winter day!





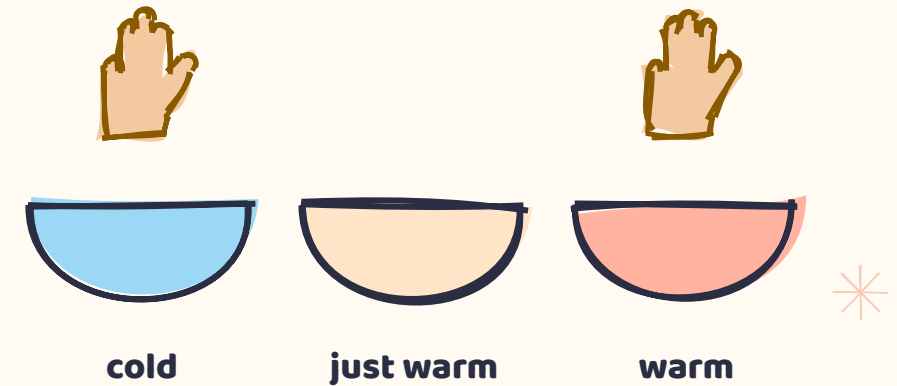
Let's wonder...

Can our hands tell the truth?



A tricky experiment

Put one hand in **cold** water and one hand in **warm** water. Wait.
Now put **both** hands in the middle bowl.
One hand says **warm**! The other says **cold**!



Our hands are confused!

Which hand is right?

Use a sensor!

Today: we build a machine that measures hot and cold with **numbers**, and shows it with a red or blue light.





Learning aims

Today I will learn to...



1



I can say if something is **hot** or **cold** using a **number**.

2



I can make a **red** or **blue** light switch on.

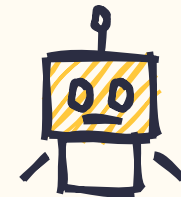
3



I can **choose the number** where red turns on.

Keywords

hot · cold · warm · cool · temperature
· degrees · sensor





Hot and cold

Why do we check temperature?



Fever

Is my little brother poorly?
Check with a thermometer



Weather

Should I wear a jumper or a
T-shirt today?



Cooking

Is the milk too hot to drink?



The fridge

Keeps food cold so it stays
fresh



Where do YOU feel hot or cold? Kitchen? Car in summer? Under a fan?

Ask at home

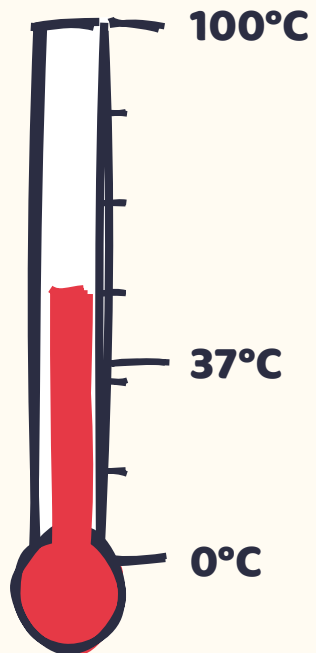
Do you have a thermometer? Where is it
kept?





Big idea

Temperature is a NUMBER



100°C boiling water (very hot!)



37°C inside our body



0°C ice

Degrees

We write °C and say degrees. Bigger number = hotter!

Try it

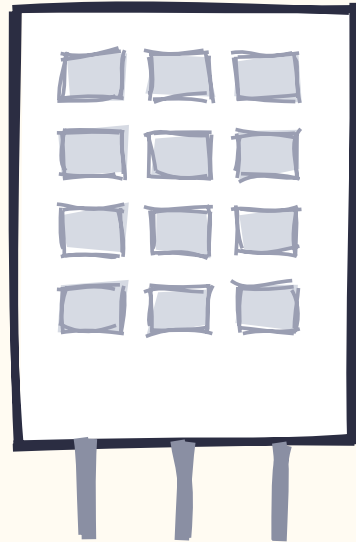
Is 30 hotter or colder than 20? What about 5?





Meet our helper

The temperature sensor



A tiny thermometer

This white box feels the air around it. It tells the Arduino the temperature as a **number**, like **27**.

Fun fact

It can also tell if the air is damp (humidity).
Like after rain!



Keep it dry!

It is not waterproof. Never put it in water or in your mouth.



Class game

Hot or cold?

How to play

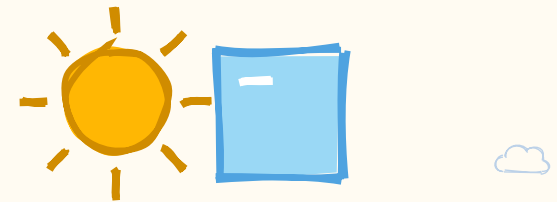
1. The teacher shows a picture card.
2. **HOT?** Stand up tall like the sun!
3. **COLD?** Curl up small like an ice cube!
4. Cards: ice cream, cup of tea, snowman, fire, the fridge, summer sun

Then discuss

Is a cup of tea **always** hot? What happens if you wait one hour?

Big idea

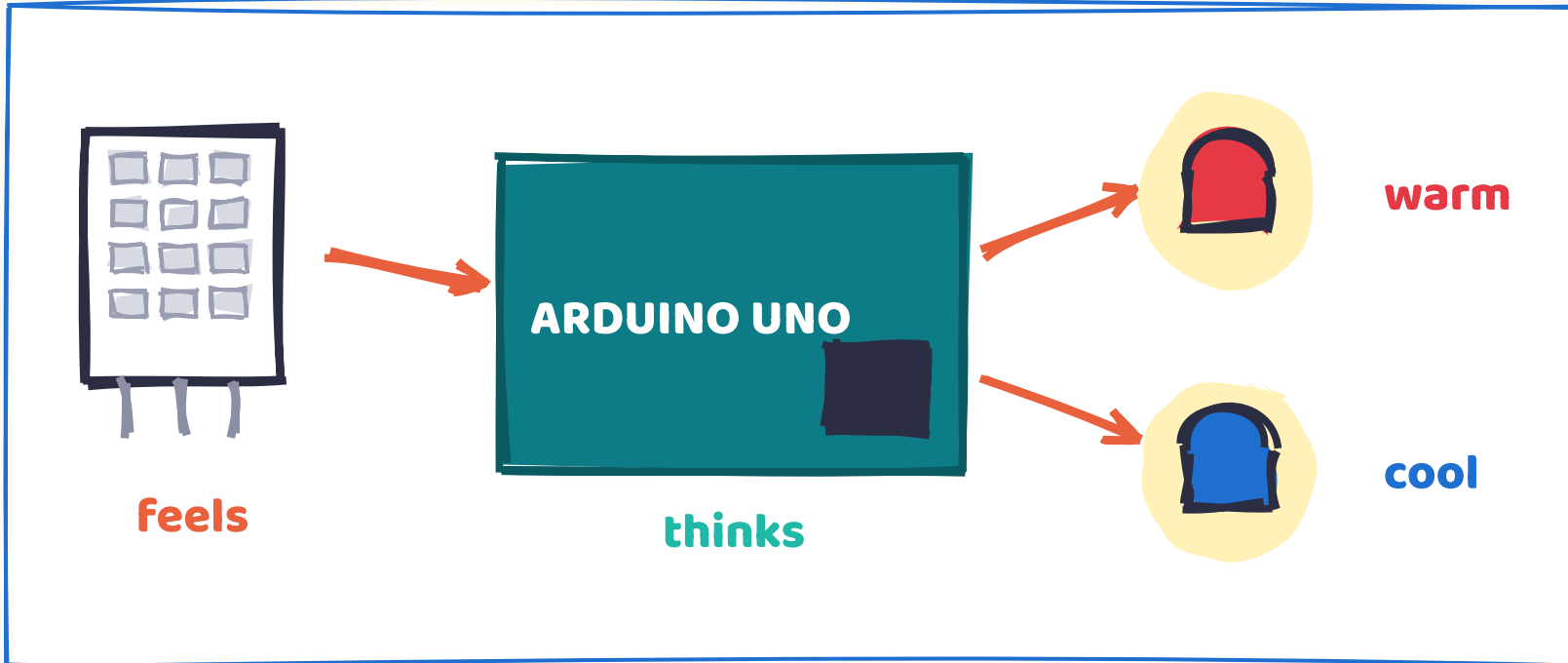
Hot things cool down and cold things warm up over time.





Our machine

Feel, think, show



How it works

The sensor **feels** the temperature.

The Arduino **thinks**: is it bigger than our number?

The lights **show** the answer.

Teacher: the sensor and two LEDs are pre-wired (data pin 2, red LED pin 12, blue LED pin 11).
Children plug in, code and test.



Time to build!

Let's code our light



when Arduino starts

forever

set temp to temperature (pin 2)

if temp > 28

turn ON red light

turn OFF blue light

else

turn ON blue light

turn OFF red light

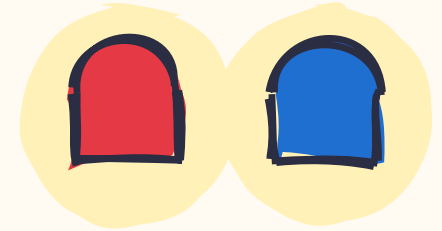
wait 2 seconds

Read it like a story

Read the temperature.

IF it is more than 28: **red**.ELSE: **blue**.

Wait, then check again!



Why wait 2 seconds?

Our sensor is a bit slow. It needs a little rest between readings.

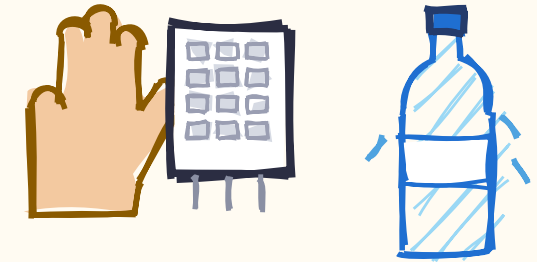




Experiment!

Make it change colour

Try this	I think...	Number	Light
Leave it on the table	red or blue?	_____	_____
Hold it gently in both hands	red or blue?	_____	_____
Hold it near a cold bottle	red or blue?	_____	_____
Blow on it	red or blue?	_____	_____



Be patient! The number changes **slowly**. Count to 20 before you look.

Cold bottle tip

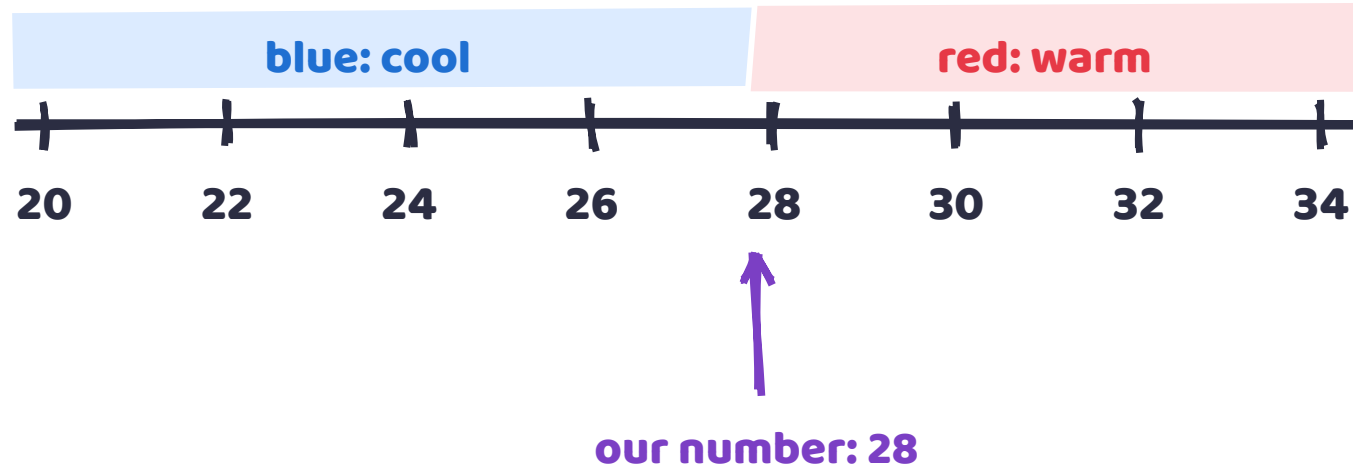
Wrap the bottle in a plastic bag so no water drips on the sensor.





You decide!

Choose YOUR hot number



Your turn

Change **28** in the code to a new number. Try **25**. Try **32**. When does the light change easily? When does it never go red?

Bigger or smaller?

If the room is 27 and your number is 25, which light shows?

Engineer word

The number where things change is called a **THRESHOLD**.





Explorers!

Where is the warmest place?



Place	Number	Light
Near the window	_____ °C	red / blue
Under the fan	_____ °C	red / blue
Near the door	_____ °C	red / blue
Inside a school bag	_____ °C	red / blue

Talk about it

Which place was **warmest**?
Which was **coolest**? Why do you think so?



Teacher

Carry the laptop with the Arduino on a tray.
Walk slowly together!





Stay safe

Hot things can hurt!



Never touch a stove, kettle, iron or hot pan.



Never put our sensor in water, tea or your mouth.



Ask a grown-up to check if food or drinks are too hot.



Cool a burn with cool running water and tell a grown-up straight away.



Remember

Our sensor measures the AIR. It is not for checking a fever.





Level up!

Hot and cold challenges



Choose your challenge



Just right: add a green light for a middle number, not too hot and not too cold.



Beep when hot: make the buzzer beep when the red light comes on.



Blink fast: make the red light blink when it is VERY hot.



Class chart: write everyone's hot number on the board. Whose was the biggest?

Think like an engineer

Why do you think a fridge has a light that switches on when you open the door?



Boss level

Make the light change at TWO numbers: blue, green, then red!





Real heroes

Thermometers keep us safe



Doctors

Check if you have a fever. 38°C or more means tell a grown-up!



Fridges

Keep food cold so germs do not grow



Farmers

Protect plants from frost and heat



If you could put a temperature light anywhere at home, **where** would you put it?

Draw it!

Draw your invention and show the class.





Show what you know

What did we learn?

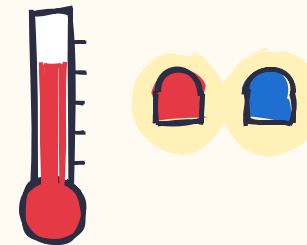
- ☒ Temperature is a number in degrees (°C)
- ☒ Bigger number = hotter
- ☒ Our sensor is a tiny thermometer for the Arduino
- ☒ IF it is hot: red. ELSE: blue
- ☒ We chose our own hot number (threshold)



Exit ticket: is 35°C hotter or colder than 20°C?

At home

With a grown-up, look at a weather app. What is today's temperature?



Next class: a
Plant Water
Alarm!

